SCHEME OF VALUATION- QP CODE: 25900314

MGU-BCA (HONOURS) REGULAR EXAMINATION OCTOBER 2025

Third SEMESTER

MG3CCRBCA200 - QUANTITATIVE TECHNIQUES

PART A

Answer any 5 questions. Each question carries 2 marks.

1. Two definitions, stubs and captions…….2 marks
   (Drawing a table and marking stubs and captions……2 marks)
2. Any one of the definition of statistics…….2 marks
3. Both are measures of variability. SD is the absolute measure of dispersion and C.V is the relative measure of dispersion…….2 marks
   or Formula for finding SD and C.V……2 marks
4. Formula $\bar{X}_w = \frac{x_1 w_1 + x_2 w_2}{w_1 + w_2}$……1 mark
   Weight of 70 = $\frac{4}{3}$…….1 mark
5. Definition of multiple regression…….1 mark
   Example……1 mark
6. Definition or correlation is perfect negative if r = -1 or drawing the graph of perfect negative correlation ……2 marks
7. Definition of independence of two events……2 marks
   or example of independence of two events……. 2 marks
8. Any one of the definitions of probability ….2 marks

PART B

Answer any 5 questions. Each question carries 6 marks.

9. Any three classification of data with examples ……6 marks
   (Each of the definition of classification- 3 marks and each of the 3 examples- 3 marks)
10. Any 4 differences between primary data and secondary data…….6 marks
    (1 ½ marks for each differences) or Definitions of both primary data and secondary data with examples (6 marks- 3 marks for each definition)
11. Definition of Quartiles with formula …2 marks
    Definition of Deciles with formula...2 marks
    Definition of percentiles with formula…2 marks
12. A) Concept or definition of mean, median and mode with simple examples…3 marks
    B) Mean =1.73(1 mark), Median=2 (1 mark) and Mode=1 (1 mark) ……3 marks
13. Taking any of the equation as y on x and the other as x on y ….1 mark
    Writing y on x equation in the form $y = a + bx$ and the x on y equation in the form $x = a + by$ ....(2 marks).

Identifying b_{yx} from the regression equation of y on x and b_{xy} from the regression equation of x on y ….1 mark

Calculating $r = \sqrt{b_{yx}\, b_{xy}}$ or $r^2 = b_{yx}\, b_{xy}$ and identifying the equation as $5x - 6y + 90 = 0$ is the regression equation of y on x and $15x - 8y - 130 = 0$ is the regression equation of x on y. ……2 marks

14. $\sum x = 599, \sum y = 1004, \sum xy = 76656$……1½marks

    $\rho_{xy} = 185.1875, \sigma_x = 12.2417, \sigma_y = 15.8508$….1½marks

    $r = \frac{\rho_{xy}}{\sigma_x \sigma_y}$ or $r = \frac{n\sum xy - (\sum x \sum y)}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}}$ and $P.E = \frac{.6745(1-r^2)}{\sqrt{n}}$…..1 mark

    $r = 0.9543\ and\ P.E = 0.0264$…….1 mark

    $r > 6\ P.E$ and significant…….1 mark.

15. P (contractor will get a plumbing contract) =P(A)= 2/3…. (1 mark)

    P (contractor will not get an electric contract) =5/9…. (1 mark)

    P (contractor will get an electric contract) =P(B)=1-5/9=4/9….. (1 mark)

    P (contractor will get at least one contract) = $P(A \cup B)$= 4/5…. (1 mark)

    $P(A \cup B) = P(A) + P(B) - P(A \cap B)$…… (1 mark)

    $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 14/45$… (1 mark)

16. Definition of simple correlation…..2 marks

    Definition of partial correlation…. 2 marks

    Definition of multiple correlation…. 2 marks

PART C

Answer any 2 questions. Each question carries 15 marks.

17. Simple bar diagram-definition, and example with figure…..5 marks

    Multiple bar diagram-definition, and example with figure…..5 marks

    Sub-divided/ Component bar diagram-definition, and example with figure…..5 marks

18. Less than cumulative frequencies 5, 13, 28, 44, 50……2 marks

    Median class 20-30, $l = 20, f = 15, m = 13$……2 marks

    $Median = l + \frac{\left(\frac{N}{2} - m\right)c}{f}$…….1 mark

    $Median = 28$ … …1 mark

    $\sum |x - median| f$: 115, 104, 45, 112, 102 total 478…….3 marks

    $Mean\ deviation = \frac{\sum |x - median| f}{N}$……2 marks

    $M.D\ about\ median = 9.56$…….1 mark

    $Coefficient\ of\ M.D = \frac{M.D}{Median}$…….2 marks

    $Coefficient\ of\ M.D = 0.3414$…….1 mark.

19. $u = x - 25, v = y - 10$

    $\sum x = 270, \sum y = 80, \sum u = 20, \sum v = -20, \sum u^2 = 326, \sum v^2 = 210, \sum uv = 15$

- 3 marks (or apply direct method)

$\rho_{uv} = \frac{\sum uv}{n} - \bar{u}\,\bar{v}$…….1 mark

$\rho_{uv} = 5.5$ ……1 mark

$b_{xy} = b_{uv} = \frac{\rho_{uv}}{\sigma_v^2}$……1 mark…….$b_{xy} = 0.3235$ ... .1 mark

$b_{yx} = b_{vu} = \frac{\rho_{uv}}{\sigma_u^2}$……1 mark…….$b_{yx} = 0.1923$…….1 mark

a) Regression equation of Y on X

$Y = \bar{Y} + b_{yx}(X - \bar{X})$……1 mark

$y = 2.8079 + .1923x$…….1 mark

Regression equation of X on Y

$X = \bar{X} + b_{xy}(Y - \bar{Y})$……1 mark

$x = 24.412 + .3235y$…….1 mark

b) When y=12; x=28.294….1 mark

c) When x=40; y=10.499….1 mark

20. Bayes Theorem Statement-5 marks

A be the event of drawing black ball,

B_1 be the event of choosing bag A, B_2 be the event of choosing bag B……3 marks

$P(B_1) = P(B_2) = 1/2$…….2 marks

$P(A/B_1) = \frac{6}{10},\ (A/B_2) = \frac{3}{7}$….2 marks

$P(B_1/A) = \frac{P(B_1)P(A/B_1)}{\sum_{i=1}^{2} P(B_i)P(A/B_i)}; i = 1,2$…….1 mark

$P(B_1/A) = \frac{7}{12}\ or\ 0.5833$…….2 marks

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